

Reference excerpt 01-006

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rected from the brain to the heart through the internal jugular veins (IJVs), epidural veins and vertebral veins (see Fig. 1). The brain-heart direction indicates that there must be a negative pressure gradient driving the blood flow (i.e. the pressure in the brain is higher than the pressure in the pressure in the right atrium (RA)). The pressure, where the IJVs begin, is the residual arterial pressure and, despite its pulsatile nature, in this study it is considered to be constant. The pressure in the RA varies according to the cardiac cycle. Its trace presents two wave peaks called a and v and two waves minima called x and y. Such waves have a precise phase relationship with the ECG waves PQRST [Applefeld(1990)], see Fig. 2. The waves a, x, v and y are also detectable at level of the neck due to the internal jugular vein (IJV) pulsation[Mackenzie(1902)]. The pressure changes generated in the RA are indeed transmitted to the IJV and affect the velocity of the blood in two ways i) modifying the pressure gradient so that it is no longer constant over time and generates a non-steady flow [Sisini et al.(2016), Kalmanson et al. (1972)], ii) cyclically varying the CSA of the IJV [Sisini et al.(2015)]. Since the walls of the IJVs are neither rigid nor collapsed, the pressure variations generated in the RA are transmitted to the IJV as a pressure wave with

nite 1 Figure 1: Geometrical model of the jugular-heart segment. The z axis is positive in the jugular-heart direction. I